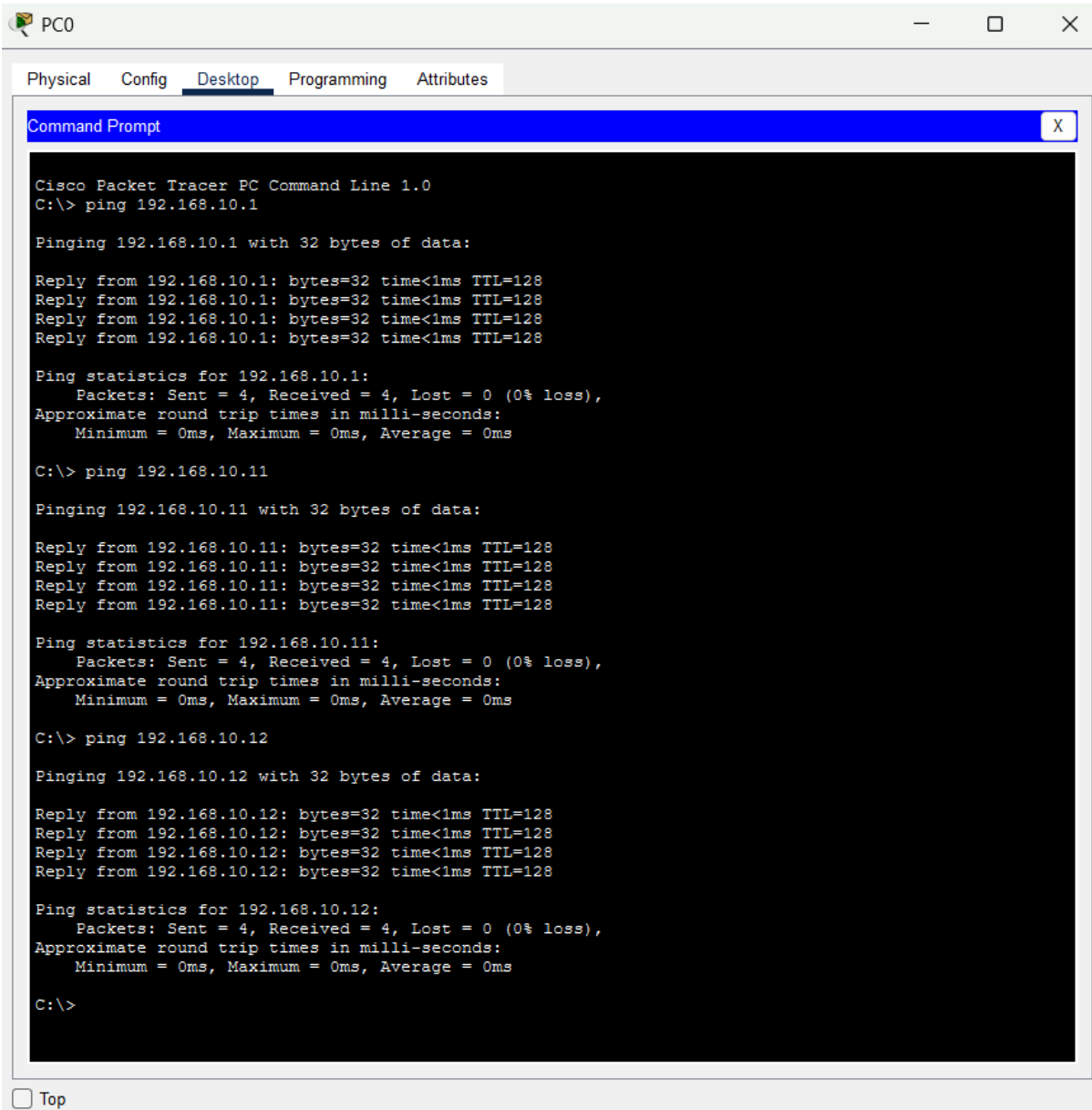
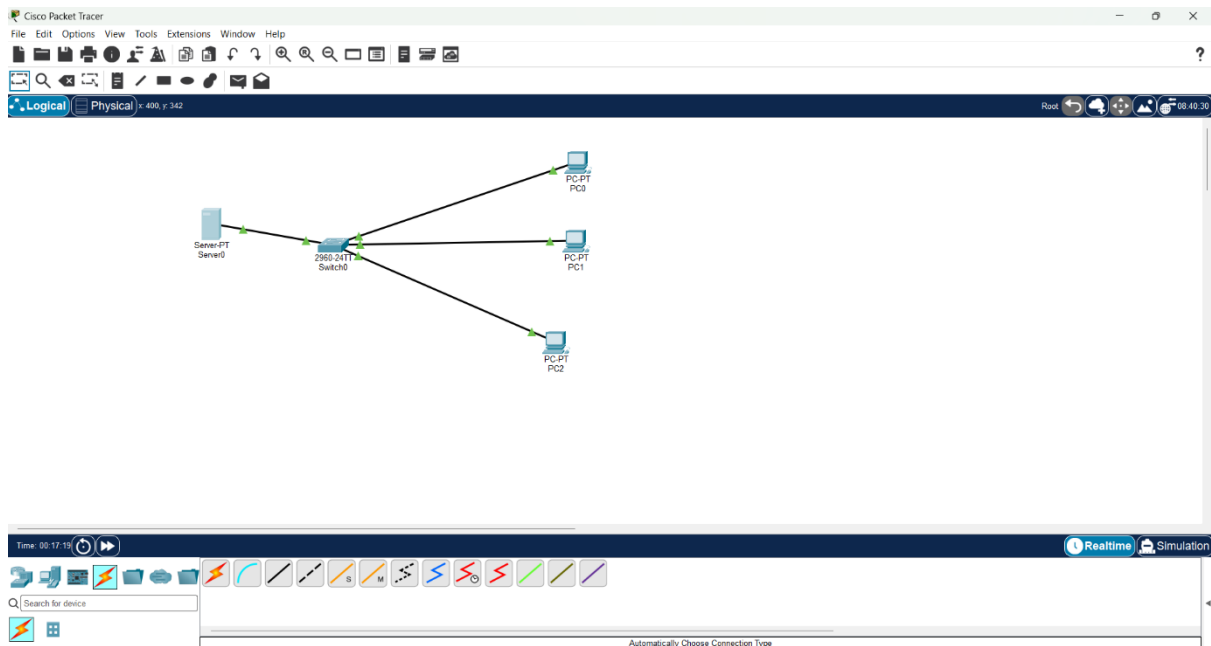
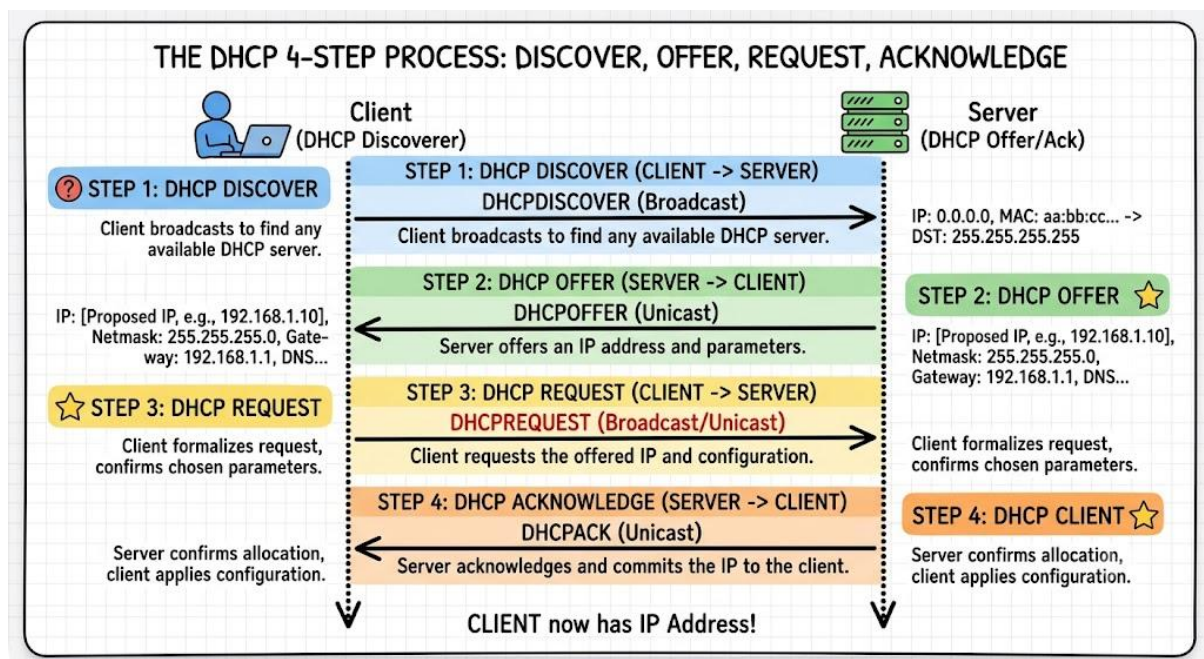


Task -1





Task-2



Task-3

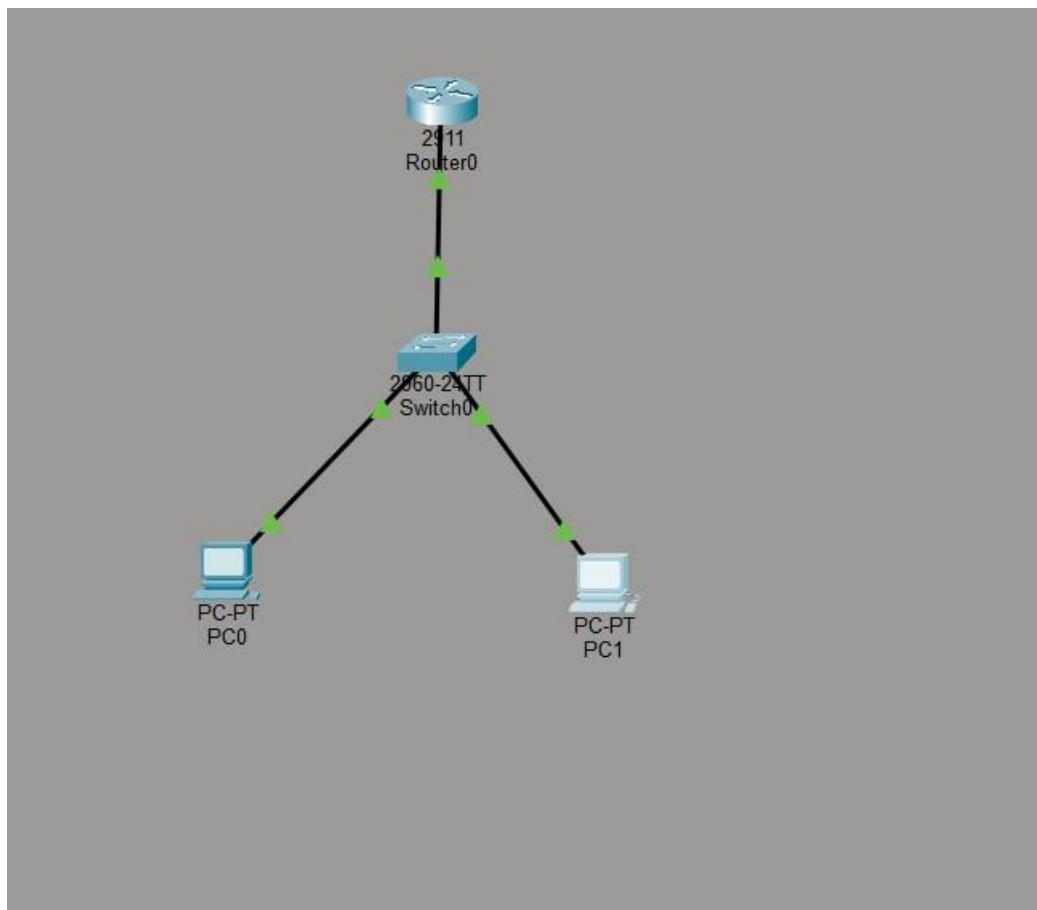
Q-Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

To configure a router as a DHCP server with only one available IP address and observe that only one client PC receives an IP address while the second PC fails to obtain one.

Requirements

- Cisco Packet Tracer
- 1 Router (2911/1941)
- 1 Switch (2960)
- 2 PCs
- Copper Straight-Through Cables

Network Topology



Router Configuration

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip dhcp excluded-address 192.168.1.3 192.168.1.254
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#show running-config
^
% Invalid input detected at '^' marker.

Router(config)#show running-config
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config
Building configuration...

Current configuration : 857 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
ip dhcp excluded-address 192.168.1.3 192.168.1.254
!
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
```

Procedure 1. Added one router, one switch, and two PCs in

Cisco Packet Tracer.

2. Connected all devices using Copper Straight-Through cables.
3. Configured the router interface with the IP address **192.168.1.1/24**.
4. Configured the router as a DHCP server.
5. Excluded all IP addresses except **192.168.1.2**, leaving only one IP available for DHCP.
6. Configured both PCs to obtain IP addresses automatically using DHCP.
7. Verified the assigned IP addresses using the **ipconfig** command.

PC0 DHCP :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

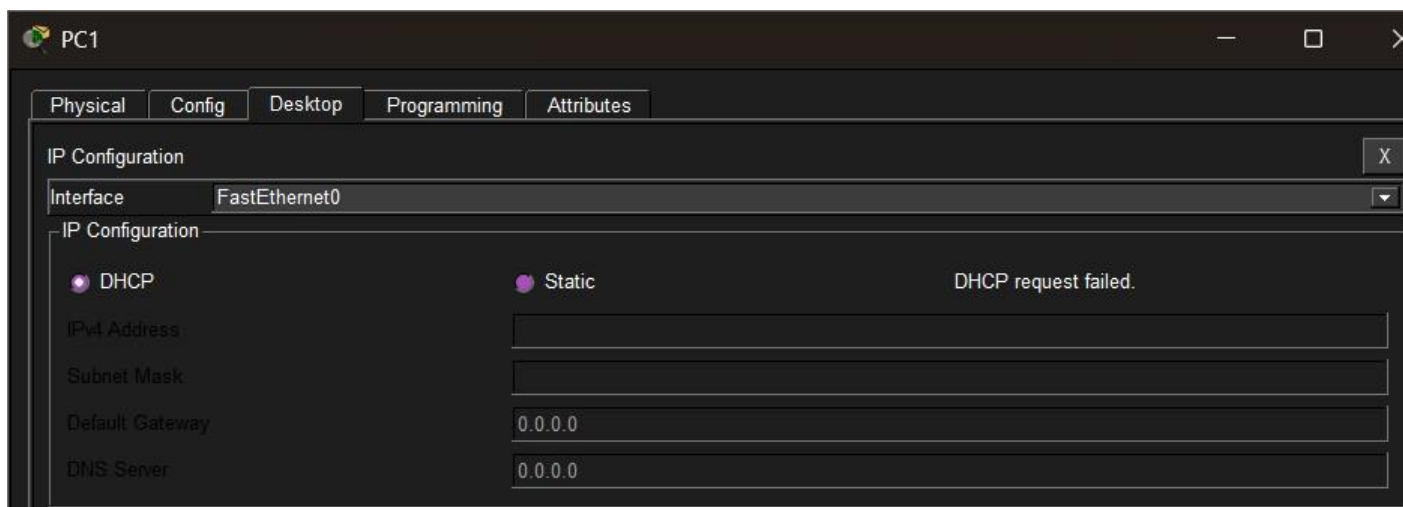
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:A3FF:FEAA:73D8
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>|
```

PC1 DHCP :



Only one IP address was available in the DHCP pool. The first PC successfully obtained that IP address through DHCP, while the second PC could not receive an IP address because the DHCP pool had no remaining addresses to assign.

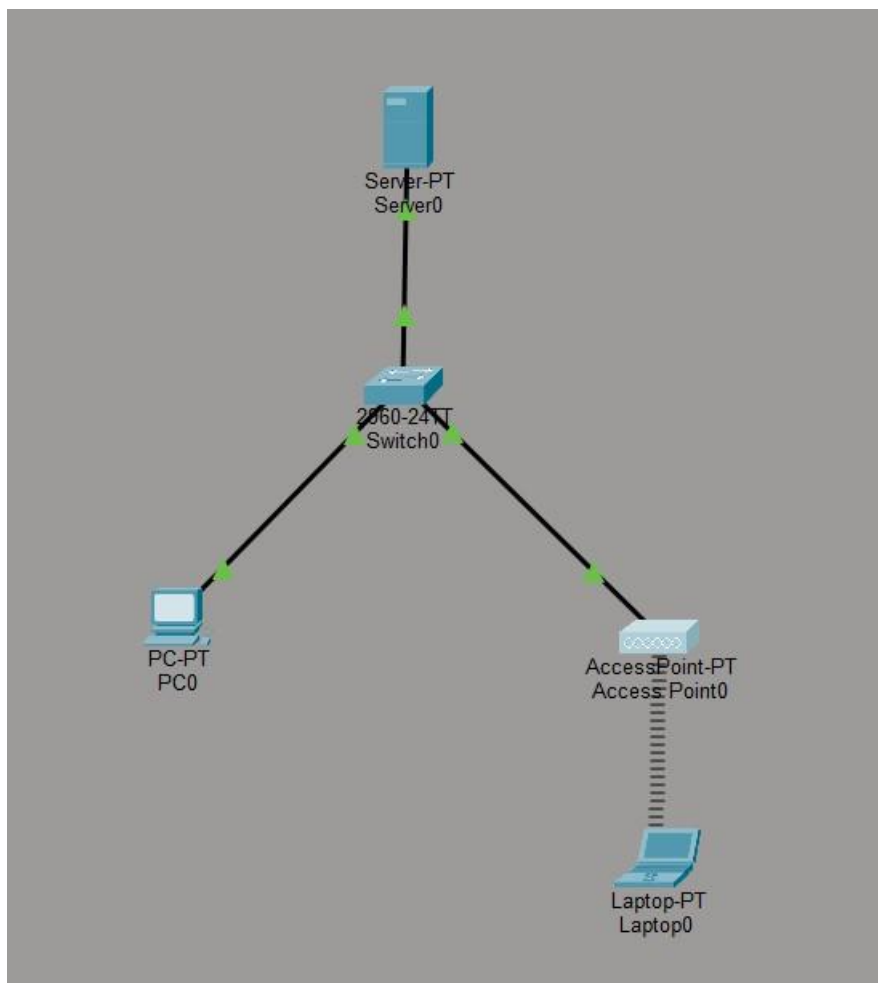
Task-4

Q- Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop

and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation

To connect a laptop wirelessly to a WRT300N Wireless Router and understand the hardware requirements for wireless communication in Cisco Packet Tracer.

Network Topology



DHCP Server Configuration

Server IP Configuration

Parameter	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0

Default Gateway	192.168.10.1
-----------------	--------------

Server0

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

192.168.10.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

DHCP Pool:

Server0

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

Start IP Address :

192

168

10

10

Subnet Mask:

255

255

255

0

Maximum Number of Users :

245

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

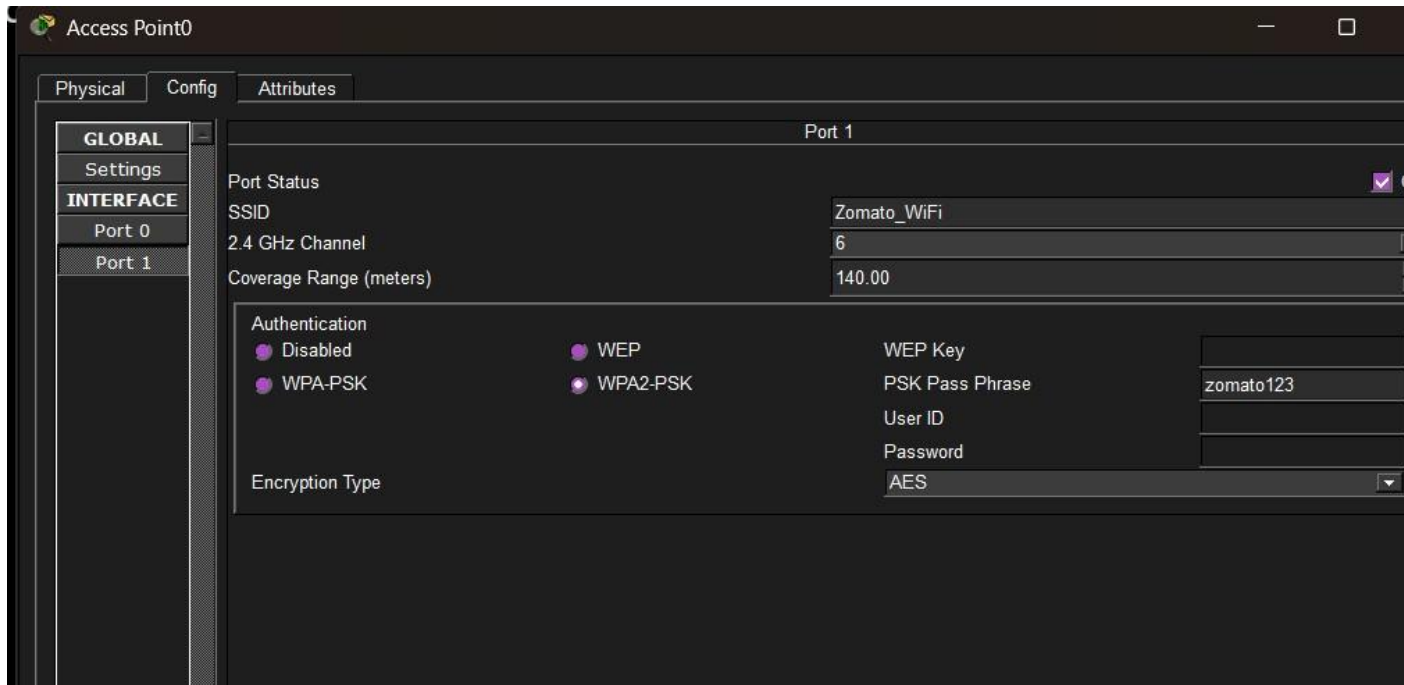
Add

Save

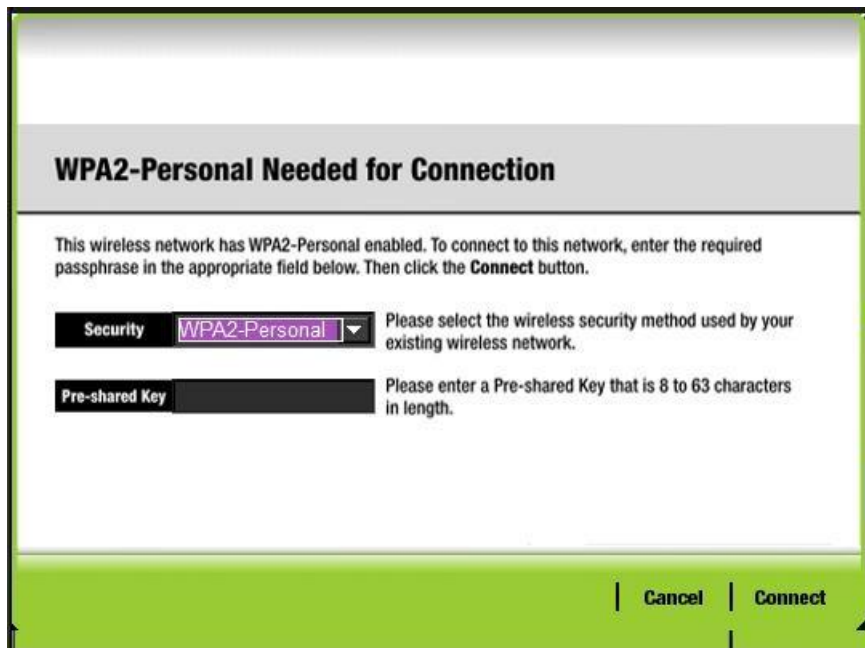
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	0.0.0.0	192.168.10.10	255.255.255.0	245	0.0.0.0	0.0.0.0

Access Point-Pt(Name and Password):



Laptop connect wireless



Laptop DHCP Ip address:

```
ipconfig

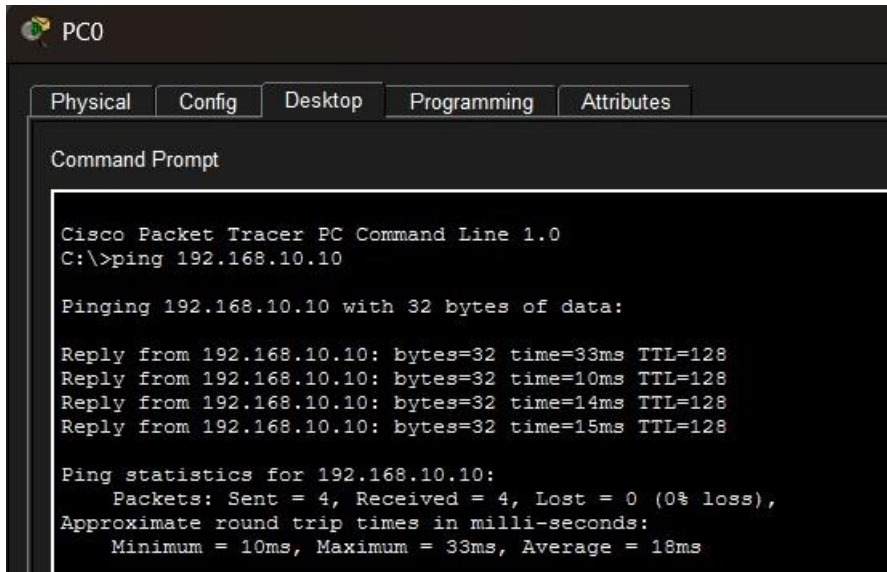
Bluetooth Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Wireless0 Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:64FF:FE22:E60A
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.10.1
```

Check connectivity:



The screenshot shows a PC0 interface with tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt window is open, displaying the output of a ping command. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=33ms TTL=128
Reply from 192.168.10.10: bytes=32 time=10ms TTL=128
Reply from 192.168.10.10: bytes=32 time=14ms TTL=128
Reply from 192.168.10.10: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 33ms, Average = 18ms
```

Why DHCP is Useful in a Restaurant WiFi Network

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices when they connect to the network.

In a restaurant such as Zomato:

- Customers connect to WiFi without manually configuring IP addresses.
 - Each device receives a unique IP address automatically.
 - Network administration becomes easier.
 - IP address conflicts are prevented.
 - New customers can join the WiFi quickly and conveniently.
-

Advantages

- Automatic IP address allocation.
- Reduces manual configuration.
- Prevents duplicate IP addresses.
- Saves time for network administrators.
- Supports multiple users efficiently.